23 Solve $5 < (x - 2)(x + 2) < 12$

You must show all your working.

$5 < x^2 - 4 < 12$

1 mark
Expand and add 4.

Add 4 throughout: $9 < x^2 < 16$

Critical values: $-4, -3, 3, 4$

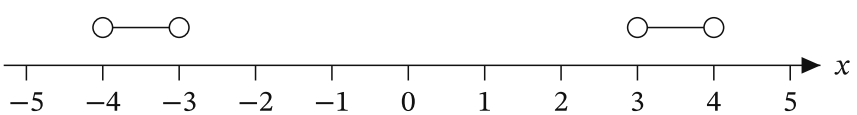
1 mark
Both square roots.

$x^2 > 9$: $x < -3$ or $x > 3$

$x^2 < 16$: $-4 < x < 4$

1 mark
Find each region.

Combine both conditions:



$-4 < x < -3$ or $3 < x < 4$

1 mark
Both open intervals.

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(Total for Question 23 is 4 marks)